Que :- A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

Code:

```
  GNU nano 6.2
#include <stdio.h>

int main() {
    int frames = 3;
    int ref[] = {1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5};
    int n = 12;

    int memory[3];
    int i, j, k, pos, max;
    int hits = 0, faults = 0;
    int found;

    // Initialize memory frames
    for(i = 0; i < frames; i++)
        memory[i] = -1;

    for(i = 0; i < n; i++) {
        found = 0;

        // Check if page is already in memory
        for(j = 0; j < frames; j++) {
            if(memory[j] == ref[i]) {
                hits++;
                found = 1;
                break;
            }
        }

        // If not found → Page Fault
        if(!found) {
            faults++;

            // Check for empty frame
            int empty = -1;
            for(j = 0; j < frames; j++) {
                if(memory[j] == -1) {
                    empty = j;
                    break;
                }
            }
```

```
GNU nano 6.2

            if(empty != -1) {
                memory[empty] = ref[i];
            } else {
                // Find optimal page to replace
                max = -1;
                pos = -1;

                for(j = 0; j < frames; j++) {
                    int next = -1;

                    for(k = i + 1; k < n; k++) {
                        if(memory[j] == ref[k]) {
                            next = k;
                            break;
                        }
                    }

                    // If page not used again
                    if(next == -1) {
                        pos = j;
                        break;
                    }

                    if(next > max) {
                        max = next;
                        pos = j;
                    }
                }

                memory[pos] = ref[i];
            }
        }
    }

    printf("Total Page Hits = %d\n", hits);
    printf("Total Page Faults = %d\n", faults);

    float hit_ratio = (float)hits / n * 100;
    float fault_ratio = (float)faults / n * 100;

    printf("Hit Percentage = %.2f%%\n", hit_ratio);
    printf("Fault Percentage = %.2f%%\n", fault_ratio);

    return 0;
}
```

OUTPUT:-

```
hari_ai@Harryfunlapcare: ~
hari_ai@Harryfunlapcare:~$ nano optimal.c
hari_ai@Harryfunlapcare:~$ gcc optimal.c -o optimal
hari_ai@Harryfunlapcare:~$ ./optimal
Total Page Hits = 5
Total Page Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%
hari_ai@Harryfunlapcare:~$
```